Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Databases 2.0**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**09.00am – 12.00pm**

Mark Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

A smart home system allows homeowners to manage various smart devices within their home. The system tracks the people, smart devices installed, the rooms where these devices are located and the interactions between users and devices.

a) Identify **FOUR (4)** entities in the smart home database system based on the scenario provided. **4**

MARK SCHEME:
Includes:

- ***resident***
- ***device***
- ***room***
- ***access log***

(1 mark per bullet point)
Any other suitable answers should be accepted.

b) The smart home database ensures that when a device is assigned to a room, the room must exist in the system. **4**
Explain in detail how this is enforced in the database and **describe** what could happen if this rule is not applied.

MARK SCHEME:

A foreign key constraint (1 mark) is applied on the device table, linking the device table to the room table (1 mark).

Award 1 mark for any of the points above, up to a max of 2 marks.

A device could be assigned to a non-existing room (1 mark), leading to data inconsistency (1 mark) where devices have no valid location in the system (1 mark).

Award 1 mark for any of the points above, up to a max of 2 marks.

Any other suitable answers should be accepted.

c) Identify TWO (2) benefits of using primary keys in the smart home database. **2**

Mark scheme:

- ***Ensures each record is unique***

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- ***Helps link data between tables (e.g. foreign keys)***
- ***Makes it easier to search and update records accurately***
- ***Prevents duplicate entries***

Any other suitable answers should be accepted.

Award 1 mark for each, up to a max of 2 marks.

Total 10 Marks

Marks

Question 2

Revisit the smart home database.

The smart home system enables homeowners to manage various smart devices, such as lights, thermostats and security cameras. The system records device interactions, such as tuning devices on or off.

Transactions must be handled correctly to maintain data consistency and avoid system errors.

a) Identify each property in **ACID**. For each ACID property, describe a transaction in the smart home database and explain how the property would apply. **8**

MARK SCHEME:

Atomicity (1 mark):
When a resident adds a new smart device, the database must either store all required details (device ID, room assignment, resident ID) or rollback if any part fails. (1 mark)

Consistency (1 mark):
If a room is deleted, all devices in that room must either be reassigned or removed to maintain consistency. (1 mark)

Isolation (1 mark):
If two residents attempt to control the same light at the same time, isolation ensures that only one command is processed at a time to avoid conflicts. (1 mark)

Durability (1 mark):
If a resident sets up a device schedule (e.g., turn on lights at 7 PM), the schedule remains saved even after a system reboot. (1 mark)

1 mark for each property, 1 mark for an example, up to a max of 8 marks.

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b) Explain the purpose of the **ROLLBACK** command in SQL. Identify a situation where it would be used. **2**

MARK SCHEME:

Purpose: The ROLLBACK command undoes all uncommitted changes in the current transaction, restoring the database to its previous state. (1 mark)
Example Use Case: If a resident accidentally removes a device but cancels the action before confirming, ROLLBACK can restore the device data instead of permanently deleting it. (1 mark)

Any suitable example should be accepted.

Total 10 marks

Marks

Question 3

10

In Question 1 and 2, you worked with a smart home system.

Now, consider a different but related system: a smart home energy monitoring system that tracks household energy usage.

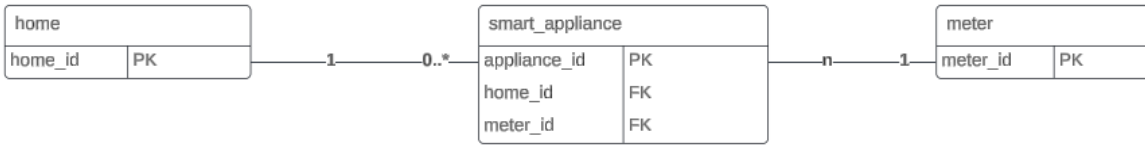
Each home contains multiple appliances and each appliance belongs to only ONE (1) home.

Energy meters are used to track electricity usage. A single meter can monitor multiple appliances, but each appliance is connected to only ONE (1) meter.

Draw an Entity Relationship Diagram for the smart home energy monitoring database.

You need to identify relationships, cardinalities and label all the primary **and** foreign keys. Clearly state which entity the keys belong to.

MARK SCHEME:



1 mark for each correctly identified entity, up to a max of 3 marks.

1 mark for each correct relationship, up to a max of 2 marks.

1 mark for each clearly defined cardinality, up to a max of 2 marks.

1 mark for all clearly labelled PKs. A compound PK is also acceptable for the smart_appliance entity type.

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1 mark for each correctly identifying FK, up to a max of 2 marks.

Total 10 marks

Marks

Question 4

a) The smart home security system allows homeowners to manage security devices and track security-related events. The system collects various details when an authorised user is added and when security devices log events. **8**

Using the information provided below, identify **EIGHT (8)** attributes that would be suitable for managing smart home security data in a database system.

Smart Home Security System Form

Date: 15 April 2025

User ID: U402

User Name: Jane Doe

Access Level: Admin

Security Devices Linked:

- **Front Door Camera**
- **Motion Sensor (Living Room)**

Security Events Recorded:

- **Front Door Camera** - **Detected Motion** (Timestamp: 10 July 2024, 3:15 AM, Severity: Medium)
- **Motion Sensor (Living Room)** - **Triggered Alarm** (Timestamp: 12 July 2024, 2:30 AM, Severity: High)

Emergency Contact Setup: Yes (Contact: +44 7890 123456)

Notifications Enabled: Yes

Device Location: Home Entrance, Living Room

Event Resolution Status: Reviewed

MARK SCHEME:
Possible attributes include

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date, user ID, user Name (first name and last name), access level, device type, device location, event type, event timestamp, event security level, event resolution status, emergency contact setup, emergency contact number, notifications enabled.

1 mark per correctly identified attribute, up to a max of 8 marks. Any other reasonable implicit attributes should be accepted. Use your discretion.

b) The smart home security database needs to store event logs efficiently. Explain why indexing would be beneficial for querying security events and given **ONE (1)** example of an attribute that should be indexed. **2**

MARK SCHEME:

Faster searching and retrieval: indexing improves query speed, allowing the system to quickly find specific security events.

Example of an attribute to index:
Event timestamp

Other acceptable indexed attributes include user id, device id, and event security level.

1 mark for an explanation and 1 mark for a suitable example.

Total 10 marks

Marks

Question 5

a) The smart home security system manages *users, security devices* and *security events*. The system performs various transactions related to these entities. **6**

Complete the CRUD matrix below to show the following transactions:

Transaction 1 – Register a new user in the security system.

Transaction 2 – Add a new security device to the system.

Transaction 3 – A user updates their personal details.

Transaction 4 – Update a security device's settings, such as its sensitivity level or recording duration.

Transaction 5 – Remove a security device that is no longer in use.

Transaction 6 – Retrieve a history of security events recorded by a specific device.

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Transaction	User	Security Device	Security Event
1			
2			
3			
4			
5			
6			

MARK SCHEME:

Award 1 mark for each completely correct row in a table, up to a maximum of 6 marks. Marks should be rounded up at the end of the question.

Transaction	User	Security Device	Security Event
1	C		
2		C	
3	U		
4		U	
5		D	D
6		R	R

b) In the smart home security database, multiple users can control multiple security devices. **4**
Describe how this relationship should be represented in a relational database.
Explain why this structure is necessary.

MARK SCHEME:

Suggested answers include, but not limited to

- ***This is a many-to-many relationship, meaning that multiple users can control multiple security devices.***

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- ***This is best structured using an intersection table that connects users to the devices they have access to.***
- ***This table establishes a proper connection between users and security devices, ensuring that access permissions are well managed.***

1 mark for each bullet point above, up to a max of 2 marks. Alternatively, suitable diagrams should be accepted and awarded up to a max of 2 marks.

- ***Without this structure, it could result in FKs at both ends each relating to the other in ways that make it unclear what the actual relationship is.***
- ***Data redundancy would occur if each device or user had to store all their access permissions directly in their own table.***
- ***The system can also store additional details, such as access levels, timestamps, or control permissions for each user-device pair.***

1 mark for each bullet point above, up to a max of 2 marks.

Any other suitable answers should be accepted.

Total 10 Marks

Marks

Question 6

Consider the smart home security system.

Write the appropriate SQL command to perform each of the following tasks to the security device data provided below.

security_device

device_id	device_type	location	status (TRUE=Active, FALSE=Inactive)
201	camera	entrance	TRUE
202	motion sensor	living room	TRUE
203	alarm system	garage	FALSE
204	camera	garage	FALSE
205	camera	living room	TRUE

a) Create the `security_device` table. **5**

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MARK SCHEME:
CREATE TABLE security_device ((1 mark)
device_id INT PRIMARY KEY, (1 mark)
device_type VARCHAR(50), (1 mark)
location VARCHAR(50), (1 mark)
status BOOLEAN (1 mark)
);

b) Update the `security_device` table to include a new column named `battery_level` to track the battery percentage of each device. The `battery_level` should be stored as an integer. **2**

MARK SCHEME:
ALTER TABLE security_device (1 mark)
ADD COLUMN battery_level INT; (1 mark)

c) The `status` of the device with `device_id` 203 needs to be updated to 'Active'. **3**

MARK SCHEME:
UPDATE security_device (1 mark)
SET status = TRUE (1 mark)
WHERE device_id = 203; (1 mark)

Total 10 marks

Marks

Question 7

The smart home security system contains the following tables.

user

user_id	first_name	last_name	contact_number
101	John	Smith	1234567
102	Aidan	Johnson	2345678
103	Sarah	Lee	3456789

security_device

device_id	device_type	location	status (TRUE=Active, FALSE=Inactive)	battery_level (%)
201	camera	entrance	TRUE	75
202	motion sensor	living room	TRUE	30
203	alarm system	garage	FALSE	50
204	camera	garage	FALSE	45
205	camera	living room	TRUE	60

access_log

log_id	access_time	user_id	device_id(FK)

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301	2025-12-23 09:00:00	101	201
302	2025-04-02 12:00:00	102	202
303	2025-10-14 08:00:00	101	203
304	2025-09-20 19:00:00	102	203
305	2025-09-16 11:00:00	102	202

a) Write an SQL command to list all access logs for devices of type “camera”. **3**

MARK SCHEME:

SELECT * FROM access_log (1 mark)

JOIN security_device ON access_log.device_id = security_device.device_id (1 mark)

WHERE security_device.device_type = 'camera'; (1 mark)

b) Write an SQL command to display each user’s name, contact number and the most recent time they accessed a security device. **4**

MARK SCHEME:

SELECT user.first_name, user.last_name, user.contact_number, (1 mark)
MAX(access_log.access_time) AS last_access_time (1 mark)
FROM access_log
JOIN user ON access_log.user_id = user.user_id (1 mark)
GROUP BY user.user_id (1 mark)

c) Write an SQL command to list all security devices where the battery level is below 50%. **3**

MARK SCHEME:
SELECT device_id (1 mark)
FROM security_device (1 mark)
WHERE battery_level < 50; (1 mark)

Any other suitable answers should be accepted.

Total 10 marks

Marks

Question 8

A smart home security installation company manages customer information, installation schedules and maintenance requests. Currently, the company relies on spreadsheets and paper-based logs to keep track of customer records, installed security devices and service appointments. **10**

As the company expands, it faces numerous challenges related to its record system. The company is now considering implementing a database system to improve efficiency.

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Identify and explain **FIVE (5)** benefits of implementing a database system over their current spreadsheet and paper-based record-keeping system.

MARK SCHEME:

Suggested answers are not limited to:

- ***Improve data accuracy***
- ***Efficient data processing/retrieval***
- ***Automatic calculation***
- ***Centralised data access for staff***
- ***Better data security***
- ***Handle large amount of data efficiently***
- ***Backup***
- ***Scalability***
- ***Automated report generation***
- ***Natural disasters recovery***

1 mark for each correctly identified advantage and 1 mark for each explanation, up to a maximum of 10 marks.

Any suitable advantages and examples should be accepted.

Total 10 marks

Marks

Question 9

a) A smart home security monitoring company provides 24/7 surveillance and remote access to security cameras, alarms and motion sensors for homeowners and businesses. The company relies on a central database to store real-time security footage, device status, access logs and customer account details. **8**

However, during a recent database failure, it was discovered that no recent backups existed and the recovery process was inadequate. This caused major disruptions in service, impacting both customers and the company's operations.

Discuss **FOUR (4)** key areas where the lack of proper database backup and recovery processes can negatively impact the smart home security monitoring company. For each area, include a specific example of how the failure could affect customers or business operations.

MARK SCHEME:

Loss of Security Footage (1 mark): Customers are unable to retrieve recorded footage of break-ins or suspicious activity. (1 mark)

Disrupted Customer Access (1 mark) Homeowners cannot disarm alarms or view security cameras remotely, leaving properties vulnerable. (1 mark)

Regulatory and Legal Risks (1 mark) A customer requests footage for legal

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evidence, but data is lost, leading to liability issues. (1 mark)

Reputation and Customer Trust Damage (1 mark) News of data loss spreads, causing customers to cancel subscriptions and switch providers. (1 mark)

1 mark for each identified key area and 1 mark for example, up to a max of 8 marks.

Any other suitable answers should be accepted.

b) A smart home installation company provides customised smart security systems for private residences. Their systems include smart locks, security cameras, motion sensors and alarm systems, all controlled through a central system. **2**

Since they deal with sensitive security data, the company wants full control over their database to ensure data privacy, security and custom system integration.

Identify a suitable cloud database model for them. Provide **ONE (1**) reason to support your answer.

MARK SCHEME:

Self-managed (1 mark).

The company needs full control over security settings to keep customer data private. (1 mark)

A self-managed database allows them to store data securely (1 mark) and customise their system without relying on an external provider. (1 mark).

1 mark for any one of the reasons above.

Any other suitable reasons should be accepted

Total 10 marks

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Marks

Question 10

A global e-commerce company is developing a backend system to manage its supply chain and logistics operations. The system must handle:

- Real-time order tracking, ensuring customers can see the status of their deliveries.
- Customer purchase history and preferences, allowing for personalised recommendations.

To meet these requirements, the company is considering adopting **NoSQL** databases for efficient scalability and performance.

a) Identify **TWO (2)** NoSQL data models that could be used to store different types of data for the platform. Provide **ONE (1)** example of data that each model would handle best. **4**

Mark scheme:

- ***Key-Value Store (1 mark): Storing real-time order tracking updates, where each order ID maps to its latest delivery status. (1 mark)***
- ***Document Store (1 mark):Managing customer profiles and purchase history, including past orders, saved addresses, and preferences. (1 mark)***

1 mark for each suitable data model, up to a maximum of 2 marks.

1 mark for each correct data, up to a maximum of 2 marks.

Any other suitable answers should be accepted.

b) Identify the **THREE (3)** elements of the 3V model. **6**

Explain how each element applies to the data challenges of the e-commerce logistics platform.

Mark scheme:

- ***Volume (1 mark):***

  The system generates and stores large amounts of data, including product inventory, order transactions, and customer records. (1 mark)

- ***Variety (1 mark):***

  The system handles different types of data, such as structured (customer details, product listings) and unstructured (customer reviews, chat logs). (1 mark)

- ***Velocity (1 mark):***

  The database processes and updates data rapidly to keep customers

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informed about orders and ensure smooth warehouse operations. (1 mark)

Any other suitable explanation should be accepted.

Total 10 marks

End of Paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1,5	Yes
2	4	Yes
3	3	Yes
4	1, 4	Yes
5	4	Yes
6	5	Yes
7	5	Yes
8	1, 2	Yes
9	1, 7	Yes
10	6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand the concepts associated with database systems	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.

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Understand the concepts associated with the relational model	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.
Understand how to design and develop a database system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Be able to develop and enhance a logical database design	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.

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Be able to develop and enhance a database system using SQL	Demonstrates little to no ability to adequately develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to develop an artefact to solve an identified issue.	Demonstrates a very good ability to develop an artefact to solve an identified issue.	Demonstrates an excellent ability to develop an artefact to solve an identified issue.
Understand the principles behind Big Data and the NoSQL databases	Displays no understanding of Big Data and NoSQL databases. Unable to recognise the key features and benefits of NoSQL over traditional databases or how Big Data impacts database design and scalability.	Shows a basic awareness of Big Data and NoSQL databases. Recognises some key aspects but fails to understand the broader implications and technical considerations involved.	Demonstrates satisfactory comprehension of Big Data principles and the operational basics of NoSQL databases. Can distinguish NoSQL databases from traditional databases and understands their use cases.	Exhibits a strong grasp of the principles behind Big Data and NoSQL databases. Can articulate the advantages of NoSQL databases in handling large datasets and different data types.	Has an in-depth understanding of Big Data and NoSQL databases. Can critically evaluate and recommend NoSQL database solutions for complex Big Data problems, demonstrating a sophisticated understanding of scalability and performance optimisation.
Understand the concepts and principles of Cloud database	Lacks understanding of the fundamental concepts and principles of Cloud databases. Cannot identify the main advantages or the operational	Has a superficial grasp of Cloud databases. Recognises some concepts but cannot articulate how Cloud databases differ from on-premises	Has a solid understanding of Cloud database concepts and principles. Understands the benefits of scalability, availability, and managed services provided by	Possesses a very good understanding of Cloud databases. Can discuss the importance of Cloud databases in modern architectures and understands	Demonstrates an excellent and comprehensive understanding of the key concepts and principles of Cloud databases. Can describe in detail how Cloud databases offer advantages in terms of accessibility, maintenance,

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	model of Cloud databases.	solutions or their benefits.	Cloud databases.	their significance in providing flexible and scalable data storage solutions.	and scalability, and can suggest appropriate use cases for Cloud databases.

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